

STAT 6440: Data Mining: Group 8

Group Assignment 3

Final source code

2026-04-17

Exploratory Analysis

In the part of our we want to explore the dataset and understand the representation of each features.

```
# load the dataset
dataLifeExpectancy <- read.csv("Life-Expectancy-Data-Updated.csv")
str(dataLifeExpectancy)

## 'data.frame':    2864 obs. of  21 variables:
## $ Country      : chr  "Turkiye" "Spain" "India" "Guyana" ...
## $ Region       : chr  "Middle East" "European Union" "Asia" "South America" ...
## $ Year         : int   2015  2015  2007  2006  2012  2006  2015  2000  2001  2008 ...
## $ Infant_deaths : num   11.1  2.7  51.5  32.8  3.4  9.8  6.6  8.7  22  15.3 ...
## $ Under_five_deaths : num   13  3.3  67.9  40.5  4.3  11.2  8.2  10.1  26.1  17.8 ...
## $ Adult_mortality : num  105.8  57.9  201.1  222.2  58 ...
## $ Alcohol_consumption : num   1.32  10.35  1.57  5.68  2.89 ...
## $ Hepatitis_B      : int   97  97  60  93  97  88  97  88  97  97 ...
## $ Measles          : int   65  94  35  74  89  86  97  99  87  92 ...
## $ BMI              : num   27.8  26  21.2  25.3  27  26.4  26.2  25.9  27.9  26.5 ...
## $ Polio            : int   97  97  67  92  94  89  97  99  97  96 ...
## $ Diphtheria       : int   97  97  64  93  94  89  97  99  99  90 ...
## $ Incidents_HIV    : num   0.08  0.09  0.13  0.79  0.08  0.16  0.08  0.08  0.13  0.43 ...
## $ GDP_per_capita   : int  11006  25742  1076  4146  33995  9110  9313  8971  3708  2235 ...
## $ Population_mln   : num   78.53  46.44  1183.21  0.75  7.91 ...
## $ Thinness_ten_nineteen_years : num   4.9  0.6  27.1  5.7  1.2  2  2.3  2.3  4  2.9 ...
## $ Thinness_five_nine_years : num   4.8  0.5  28  5.5  1.1  1.9  2.3  2.3  3.9  3.1 ...
## $ Schooling        : num   7.8  9.7  5  7.9  12.8  7.9  12  10.2  9.6  10.9 ...
## $ Economy_status_Developed : int   0  1  0  0  1  0  0  1  0  0 ...
## $ Economy_status_Developing : int   1  0  1  1  0  1  1  0  1  1 ...
## $ Life_expectancy   : num   76.5  82.8  65.4  67  81.7  78.2  71.2  71.2  71.9  68.7 ...

print(dim(dataLifeExpectancy))

## [1] 2864    21
```

Data characteristics

In we want to understand the information each feature in the data set and capture missing information in data. From the data, we observed the data was cleaned and there were no missing values. The total number of observation collected are 2864 and 21 individual features including the response variable.

```
colSums(is.na(dataLifeExpectancy))
```

```
##          Country          Region
##          0              0
##      Year      Infant_deaths
##          0              0
##      Under_five_deaths      Adult_mortality
##          0              0
##      Alcohol_consumption      Hepatitis_B
##          0              0
##      Measles              BMI
##          0              0
##      Polio              Diphtheria
##          0              0
##      Incidents_HIV      GDP_per_capita
##          0              0
##      Population_mln Thinness_ten_nineteen_years
##          0              0
##      Thinness_five_nine_years      Schooling
##          0              0
##      Economy_status_Developed      Economy_status_Developing
##          0              0
##      Life_expectancy
##          0
```

```
head(dataLifeExpectancy)
```

```
##      Country          Region Year Infant_deaths Under_five_deaths
## 1  Turkiye      Middle East 2015      11.1      13.0
## 2   Spain    European Union 2015       2.7       3.3
## 3   India        Asia 2007      51.5      67.9
## 4  Guyana    South America 2006      32.8      40.5
## 5  Israel      Middle East 2012       3.4       4.3
## 6 Costa Rica Central America and Caribbean 2006       9.8      11.2
##      Adult_mortality Alcohol_consumption Hepatitis_B Measles  BMI Polio Diphtheria
## 1      105.8240      1.32      97      65 27.8      97      97
## 2       57.9025     10.35      97      94 26.0      97      97
## 3      201.0765      1.57      60      35 21.2      67      64
## 4      222.1965      5.68      93      74 25.3      92      93
## 5       57.9510      2.89      97      89 27.0      94      94
## 6       95.2200      4.19      88      86 26.4      89      89
##      Incidents_HIV GDP_per_capita Population_mln Thinness_ten_nineteen_years
## 1       0.08      11006      78.53      4.9
## 2       0.09     25742      46.44      0.6
## 3       0.13      1076     1183.21     27.1
## 4       0.79      4146       0.75      5.7
## 5       0.08     33995       7.91      1.2
## 6       0.16      9110       4.35      2.0
##      Thinness_five_nine_years Schooling Economy_status_Developed
## 1       4.8       7.8      0
## 2       0.5       9.7      1
## 3      28.0       5.0      0
```

```
## 4          5.5      7.9          0
## 5          1.1     12.8          1
## 6          1.9      7.9          0
## Economy_status_Developing Life_expectancy
## 1          1        76.5
## 2          0        82.8
## 3          1        65.4
## 4          1        67.0
## 5          0        81.7
## 6          1        78.2
```

- This data were collected from various continents in the world showing the occurrence of death. The table shows the number of observations collected from each continent. From the data we observed that majority of observations were collected from Africa representing 816 and least collected is North America with an amount of 48.

```
table(dataLifeExpectancy$Region)
```

```
##
##          Africa          Asia
##          816          432
## Central America and Caribbean European Union
##          304          432
##          Middle East      North America
##          224          48
##          Oceania          Rest of Europe
##          176          240
##          South America
##          192
```

```
# extract the features of interest
life_expectancy_num_vars <- dataLifeExpectancy %>%
  select(where(is.numeric)) %>%
  select(-Year)
```

- We want to have the summary statistic of country and their life expectancy. Extracted few of the features to under summary information under each continents using ; mortality information both infants and adult.

From the data we observed that Africa has the lowest median life expectancy among all the continent with a value of 57.05 follow by Oceania and Asia.

- Interestingly, Africa has highest infants mortality follow by the Asia continent
- We only have to two categorical features ; Economy_status_Developed and Economy_status_Developing. This was generated based the Status from the original datasets.

```
summary(life_expectancy_num_vars)
```

```
## Infant_deaths    Under_five_deaths Adult_mortality  Alcohol_consumption
## Min.      : 1.80    Min.      : 2.300    Min.      : 49.38    Min.      : 0.000
```

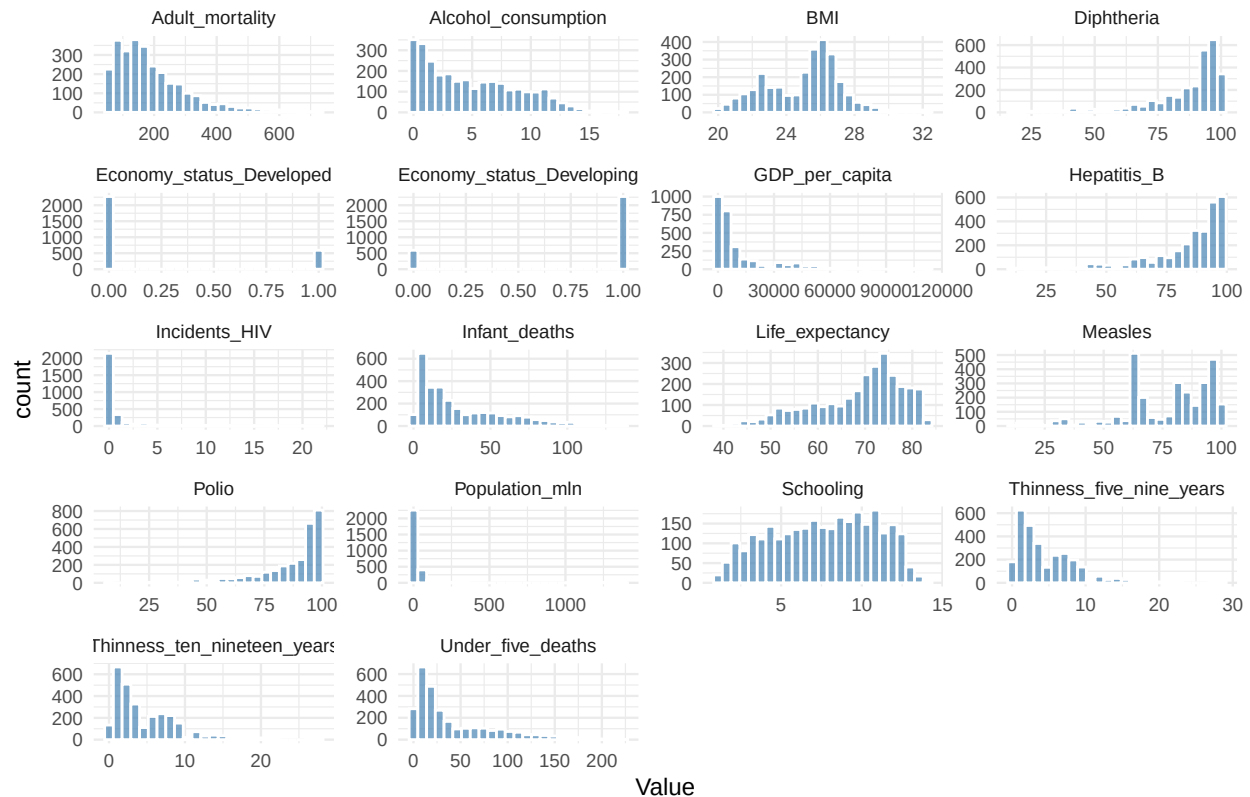
```
## 1st Qu.: 8.10    1st Qu.: 9.675    1st Qu.:106.91    1st Qu.: 1.200
## Median : 19.60    Median : 23.100    Median :163.84    Median : 4.020
## Mean : 30.36    Mean : 42.938    Mean :192.25    Mean : 4.821
## 3rd Qu.: 47.35    3rd Qu.: 66.000    3rd Qu.:246.79    3rd Qu.: 7.777
## Max. :138.10    Max. :224.900    Max. :719.36    Max. :17.870
## Hepatitis_B    Measles    BMI    Polio    Diphtheria
## Min. :12.00    Min. :10.00    Min. :19.80    Min. : 8.0    Min. :16.00
## 1st Qu.:78.00    1st Qu.:64.00    1st Qu.:23.20    1st Qu.:81.0    1st Qu.:81.00
## Median :89.00    Median :83.00    Median :25.50    Median :93.0    Median :93.00
## Mean :84.29    Mean :77.34    Mean :25.03    Mean :86.5    Mean :86.27
## 3rd Qu.:96.00    3rd Qu.:93.00    3rd Qu.:26.40    3rd Qu.:97.0    3rd Qu.:97.00
## Max. :99.00    Max. :99.00    Max. :32.10    Max. :99.0    Max. :99.00
## Incidents_HIV    GDP_per_capita    Population_mln
## Min. : 0.0100    Min. : 148    Min. : 0.080
## 1st Qu.: 0.0800    1st Qu.: 1416    1st Qu.: 2.098
## Median : 0.1500    Median : 4217    Median : 7.850
## Mean : 0.8943    Mean : 11541    Mean : 36.676
## 3rd Qu.: 0.4600    3rd Qu.: 12557    3rd Qu.: 23.688
## Max. :21.6800    Max. :112418    Max. :1379.860
## Thinness_ten_nineteen_years    Thinness_five_nine_years    Schooling
## Min. : 0.100    Min. : 0.1    Min. : 1.100
## 1st Qu.: 1.600    1st Qu.: 1.6    1st Qu.: 5.100
## Median : 3.300    Median : 3.4    Median : 7.800
## Mean : 4.866    Mean : 4.9    Mean : 7.632
## 3rd Qu.: 7.200    3rd Qu.: 7.3    3rd Qu.:10.300
## Max. :27.700    Max. :28.6    Max. :14.100
## Economy_status_Developed    Economy_status_Developing    Life_expectancy
## Min. :0.0000    Min. :0.0000    Min. :39.40
## 1st Qu.:0.0000    1st Qu.:1.0000    1st Qu.:62.70
## Median :0.0000    Median :1.0000    Median :71.40
## Mean :0.2067    Mean :0.7933    Mean :68.86
## 3rd Qu.:0.0000    3rd Qu.:1.0000    3rd Qu.:75.40
## Max. :1.0000    Max. :1.0000    Max. :83.80
```

- Distribution of each numeric features. We wanted to understand the skewness of the distribution under each feature of the dataset. We observed that most of the features are righted and lefted skewed in nature

```
# extract the features of interest
num_vars <- dataLifeExpectancy %>%
  select(where(is.numeric)) %>%
  select(-Year)

# All histograms at once
num_vars %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Value)) +
  geom_histogram(bins = 25, fill = "steelblue", color = "white", alpha = 0.7) +
  facet_wrap(~ Variable, scales = "free", ncol = 4) +
  labs(title = "Distributions of All Numeric Features") +
  theme_minimal(base_size = 9)
```

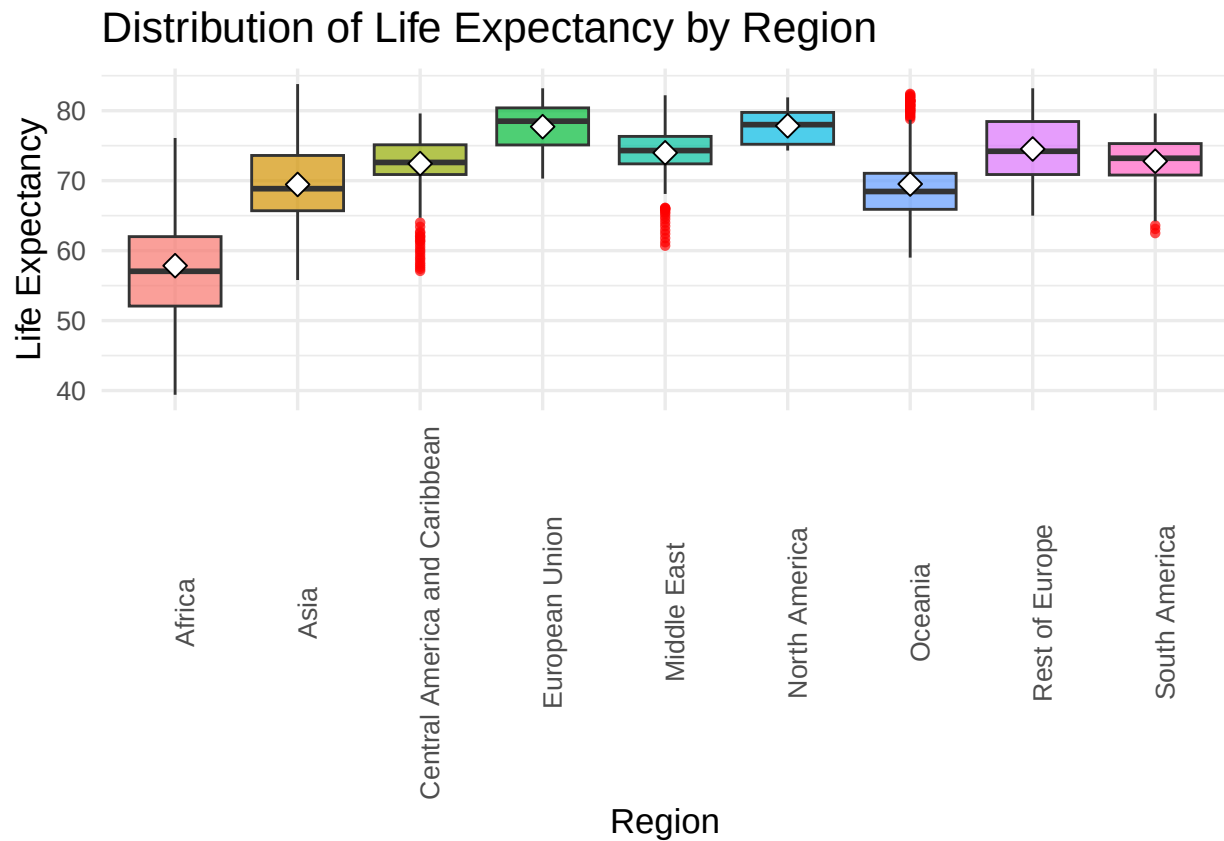
Distributions of All Numeric Features



In this part of our analysis, we want to understand the distribution of the life expectancy by region. We observed some level of outlying points for some of the features using box and whisker plot.

```
#
ggplot(dataLifeExpectancy, aes(x = Region, y = Life_expectancy, fill = Region)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 16, alpha = 0.7) +
  stat_summary(fun = mean, geom = "point", shape = 23,
              size = 3, fill = "white", color = "black") + # mean diamond

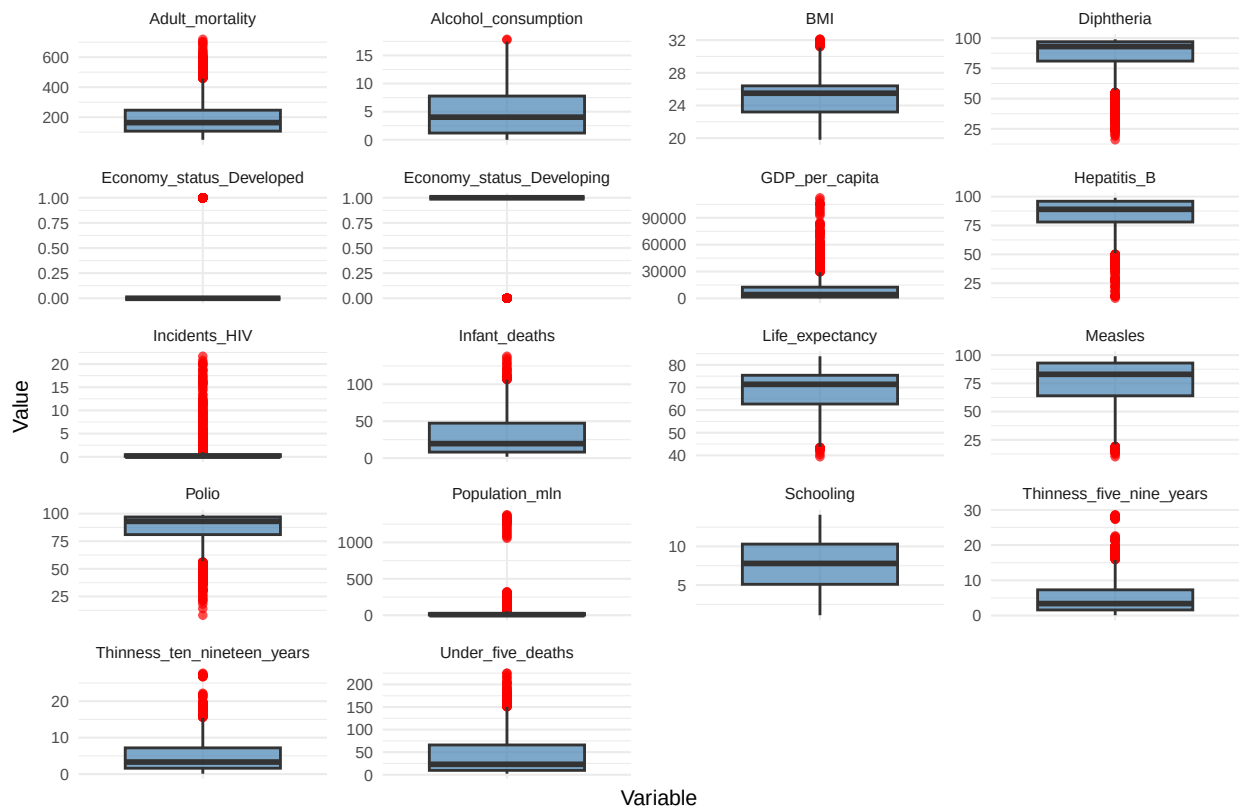
labs(
  title = "Distribution of Life Expectancy by Region",
  x = "Region",
  y = "Life Expectancy",
  fill = "Region"
) +
theme_minimal(base_size = 13) +
theme(axis.text.x = element_text(angle = 90, hjust = 0.5)) +
theme(legend.position = "none")
```



- Individual distribution without considering the region.

```
num_vars %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Variable, y = Value)) +
  geom_boxplot(fill = "steelblue", outlier.colour = "red",
               outlier.size = 1, alpha = 0.7) +
  facet_wrap(~ Variable, scales = "free", ncol = 4) +
  labs(title = "Outlier Detection Across Numeric Variables") +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_blank())
```

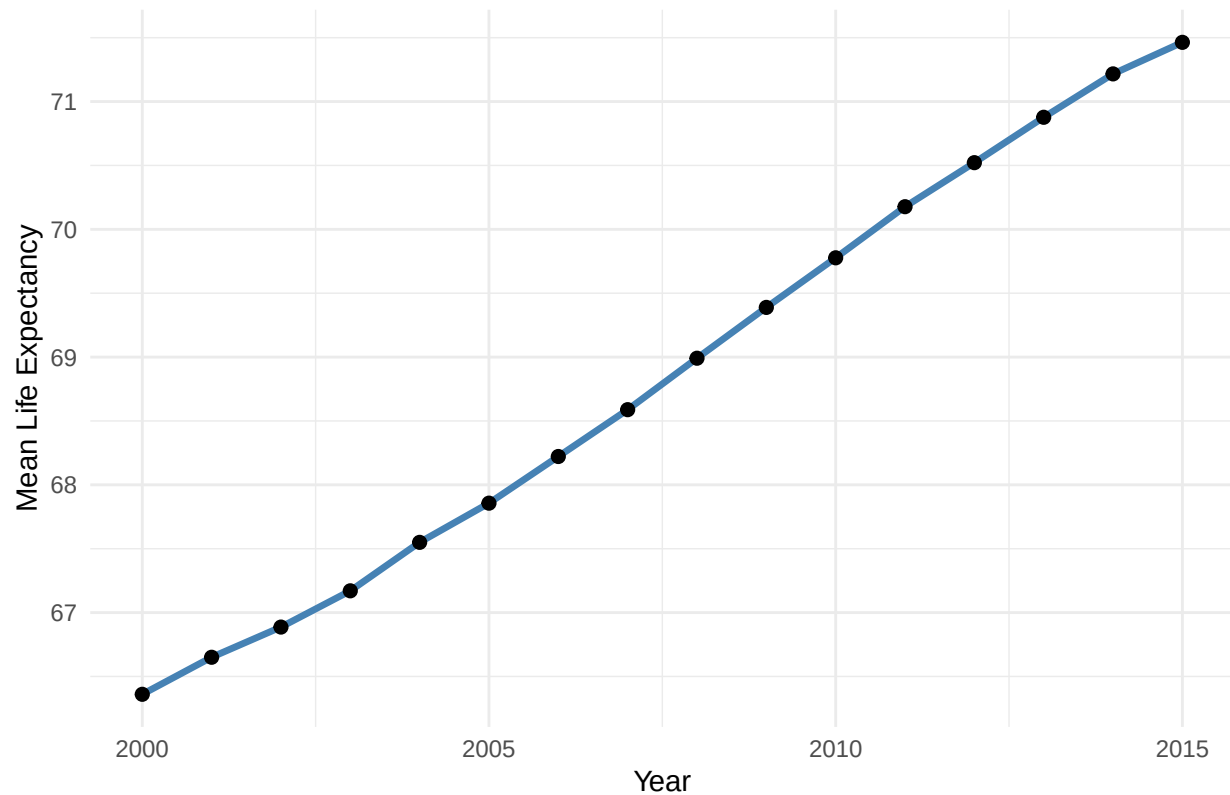
Outlier Detection Across Numeric Variables



- This is trend of average life expectancy over the years combining all information from the regions. We wanted to group by the years and under the trends of life expectancy over years. We observed that there is an increase in life expectancy over years combined information.

```
dataLifeExpectancy %>%
  group_by(Year) %>%
  summarise(mean_le = mean(Life_expectancy, na.rm = TRUE)) %>%
  ggplot(aes(x = Year, y = mean_le)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_point(size = 2) +
  labs(title = "Average Life Expectancy Over Time",
       x = "Year", y = "Mean Life Expectancy") +
  theme_minimal()
```

Average Life Expectancy Over Time



```
# extract the numeric information
num_vars <- dataLifeExpectancy %>%
  select(where(is.numeric)) %>%
  select(-Year)

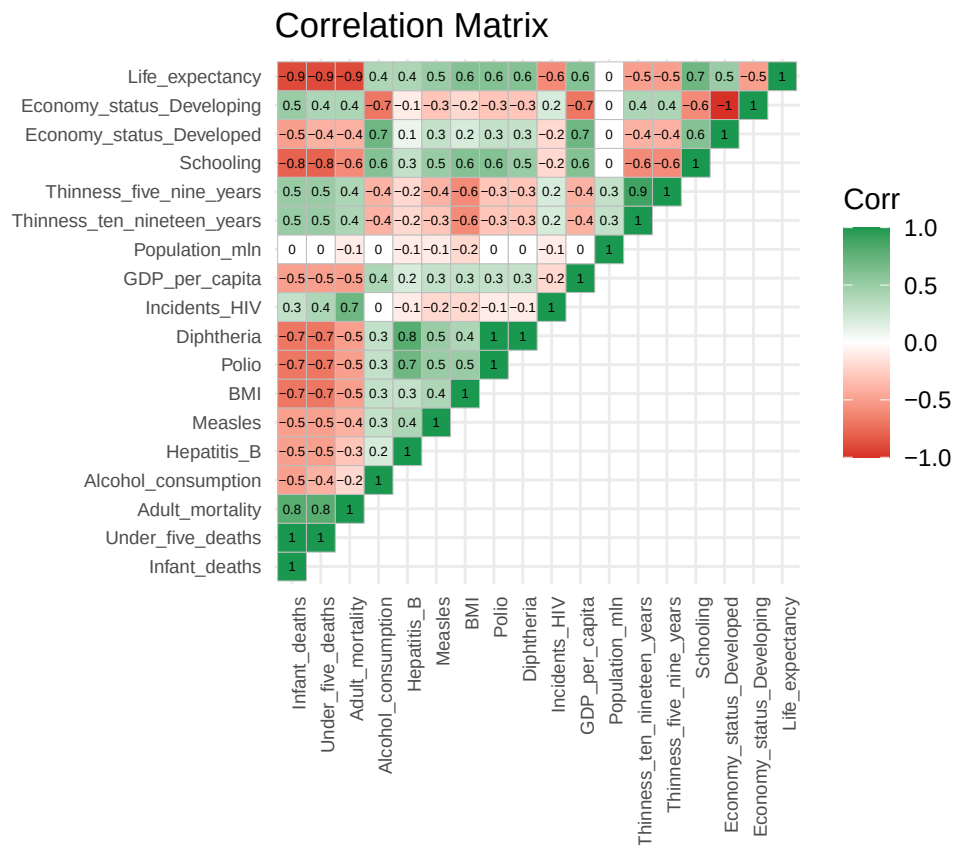
cor_matrix <- cor(num_vars, use = "pairwise.complete.obs")

# Correlation with target only
cor_with_target <- sort(cor_matrix[, "Life_expectancy"], decreasing = TRUE)
print(cor_with_target)
```

```
##           Life_expectancy           Schooling
##           1.000000000           0.73248447
##           Polio           Diphtheria
##           0.64121746           0.62754139
##           BMI           GDP_per_capita
##           0.59842332           0.58308972
## Economy_status_Developed           Measles
##           0.52379098           0.49001859
##           Hepatitis_B           Alcohol_consumption
##           0.41780443           0.39915911
##           Population_mln           Thinness_five_nine_years
##           0.02629788           -0.45816623
## Thinness_ten_nineteen_years           Economy_status_Developing
##           -0.46782450           -0.52379098
##           Incidents_HIV           Infant_deaths
```

```
##           -0.55302746           -0.92003192
##      Under_five_deaths      Adult_mortality
##           -0.92041913           -0.94536036
```

```
# Full heatmap
ggcorrplot(round(cor_matrix,1),
  method = "square",
  type = "upper",
  lab = TRUE,
  show.diag = TRUE,
  lab_size = 1.9,
  colors = c("#d73027", "white", "#1a9850"),
  title = "Correlation Matrix") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7),
    axis.text.y = element_text(size = 7))
```



```
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##      group_rows
```

Table 1: Correlation Matrix — WHO Life Expectancy

	Infant_00deaths	Under_5deaths	Adult_mortality	Hepatitis_B	Polio	Diphtheria	GDP_per_capita	Population_mln	Thinness_15-19	Thinness_5-14	Schooling	Economy_status	Life_expectancy
Infant_00deaths	1.00	0.79	-	-	-	-	0.35	-	0.01	0.49	0.48	-	0.48
Under_5deaths	0.45	1.00	0.51	0.53	0.66	0.74	0.72	0.51	-	0.79	0.48	-	0.92
Adult_mortality	0.41	0.51	1.00	0.51	0.67	0.74	0.73	0.47	0.01	0.77	0.43	-	0.92
Alcohol_consumption	0.24	0.34	0.42	0.52	0.52	0.51	0.51	0.05	-	0.58	0.43	-	0.95
Hepatitis_B	0.45	0.41	0.24	1.00	0.32	0.28	0.30	0.30	0.03	0.04	0.45	0.43	0.67
Measles	0.51	0.51	0.34	0.17	1.00	0.43	0.35	0.72	0.08	0.08	0.21	0.21	0.11
BMI	0.53	0.51	0.42	0.32	0.43	1.00	0.42	0.51	0.15	0.10	0.34	0.37	0.30
Polio	0.66	0.67	0.52	0.28	0.35	0.42	1.00	0.46	0.16	0.17	0.60	0.60	0.24
Diphtheria	0.74	0.74	0.52	0.30	0.72	0.51	0.46	1.00	0.15	0.03	0.31	0.31	0.28
Incidents_HIV	0.72	0.73	0.51	0.30	0.76	0.49	0.43	0.95	0.15	0.03	0.30	0.30	0.29
GDP_per_capita	0.03	0.08	0.15	0.16	0.15	0.15	1.00	-	0.17	0.06	0.20	0.18	0.55
Population_mln	0.44	0.16	0.31	0.34	0.31	0.31	-	1.00	0.17	0.04	0.38	0.38	0.67
Thinness_15-19	0.01	0.05	0.04	0.08	0.10	0.17	0.03	0.03	1.00	0.26	1.00	0.94	0.42
Thinness_5-14	0.45	0.21	0.34	0.60	0.31	0.30	0.38	0.38	0.26	1.00	0.94	1.00	0.41
Schooling	0.43	0.21	0.37	0.60	0.31	0.30	0.38	0.38	0.26	0.94	1.00	1.00	0.73
Economy_status_Developed	0.79	0.77	0.58	0.62	0.35	0.50	0.64	0.55	0.58	-	-	-	0.60
Economy_status_Developing	0.48	0.43	0.43	0.30	0.24	0.28	0.29	0.29	0.67	-	-	-	0.52
Life_expectancy	0.92	0.92	0.95	0.67	0.11	0.30	0.24	0.28	0.67	0.60	1.00	1.00	0.52

```
cor_df <- as.data.frame(round(cor_matrix, 2))
```

```
cor_df %>%
  kbl(caption = "Correlation Matrix - WHO Life Expectancy") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = TRUE, font_size = 11) %>%
  column_spec(1, bold = TRUE) %>%
  row_spec(0, bold = TRUE, background = "#f5f5f5")
```

```
## Warning in latex_new_row_builder(target_row, table_info, bold, italic,
## monospace, : Setting full_width = TRUE will turn the table into a tabu
## environment where colors are not really easily configurable with this package.
## Please consider turn off full_width.
```

From the datasets, we observation the features were cleaned without missing values.

Ensemble models implementation

In this session, we will extract all the numeric information and after we will split the data into train and test. We will fit with the train set and assess the performance of the best model selected.

```
# setting a seed
RNGkind (sample.kind = "Rounding")

## Warning in RNGkind(sample.kind = "Rounding"): non-uniform 'Rounding' sampler
## used

set.seed(0) # set a seed
p2_Expectancy <- partition.2(num_vars, 0.8)
training_dataExpectancy <- p2_Expectancy$data.train
test_dataExpectancy <- p2_Expectancy$data.test

cat("Dimension of the train set: ", dim(training_dataExpectancy), "\n")

## Dimension of the train set:  2291 18

cat("Dimension of the test set: ", dim(test_dataExpectancy))

## Dimension of the test set:  573 18
```

We will implement the model implementation pipeline.

```
# set the cross-validation
ctrl <- trainControl(
  method      = "cv",
  number      = 5,          # 5-fold CV
  verboseIter = FALSE,
  allowParallel = TRUE
)
```

Bagging In this case, we perform 5-folds cross-validation

```
# time the bagging
time_bag <- system.time({
  set.seed(123)
  model_bag <- train(
    Life_expectancy ~ .,
    data      = training_dataExpectancy,
    method    = "treebag",
    trControl = ctrl,
    nbagg     = 50
  )
})

cat("Bagging done\n")

## Bagging done
```

```
print(model_bag)
```

```
## Bagged CART
##
## 2291 samples
## 17 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 1833, 1833, 1833, 1831, 1834
## Resampling results:
##
## RMSE      Rsquared    MAE
## 2.250614  0.9425079  1.785548
```

```
print(time_bag)
```

```
## user system elapsed
## 1.393 0.015 1.415
```

```
time_rf <- system.time({
set.seed(123)
model_rf <- train(
  Life_expectancy ~ .,
  data      = training_dataExpectancy,
  method    = "rf",
  trControl = ctrl,
  tuneGrid  = expand.grid(mtry = c(3, 5, 7, 9)),
  ntree     = 500,
  importance = TRUE
)})
cat("Random Forest done\n")
```

Random Forest

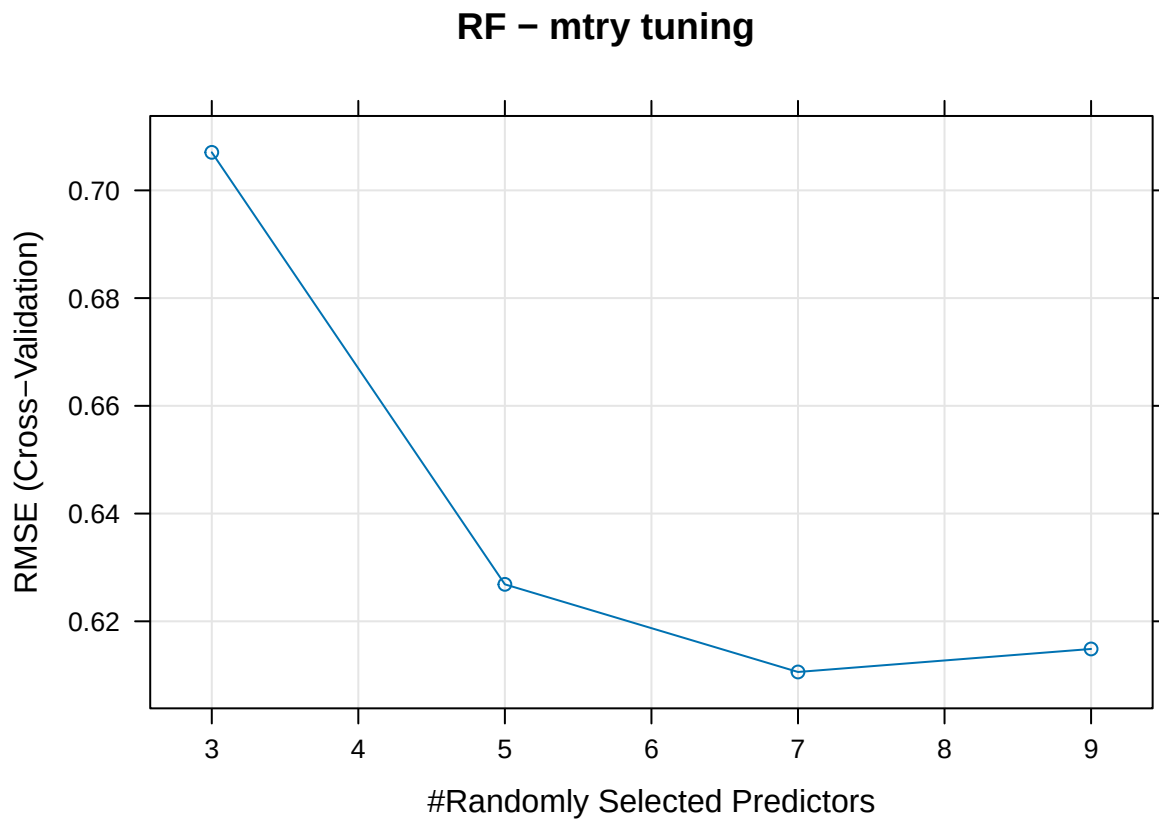
```
## Random Forest done
```

```
print(model_rf)
```

```
## Random Forest
##
## 2291 samples
## 17 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 1833, 1833, 1833, 1831, 1834
```

```
## Resampling results across tuning parameters:
##
##   mtry  RMSE      Rsquared  MAE
##   3     0.7070573  0.9947549  0.4919411
##   5     0.6268641  0.9957030  0.4444211
##   7     0.6105974  0.9958478  0.4362600
##   9     0.6148662  0.9957412  0.4381663
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was mtry = 7.
```

```
plot(model_rf, main = "RF - mtry tuning")
```



```
print(time_rf)
```

```
##   user  system elapsed
## 63.744   0.224  64.060
```

Adaboost We had to perform hyperparameter tuning for number of boosting interactions, max tree depth and learning rate.

```
# AdaBoost in caret uses method = "adaboost"
# Requires: install.packages("fastAdaboost")
tuneGrid <- expand.grid(
```

```

n.trees = c(50, 100, 150), # Number of boosting iterations
interaction.depth = c(1, 2, 3), # Max tree depth (1 = stumps)
shrinkage = c(0.1, 0.2), # Learning rate (smaller = slower but more robust)
n.minobsinnode = c(10, 20)
)

time_ada <- system.time({
set.seed(123)
modelLookup("gbm")
model_ada <- train(
  Life_expectancy ~ .,
  data = training_dataExpectancy,
  method = "gbm",
  trControl = ctrl,
  tuneGrid = tuneGrid,
  verbose = FALSE
)})

cat("AdaBoost done\n")

```

```
## AdaBoost done
```

```
print(model_ada)
```

```
## Stochastic Gradient Boosting
```

```
##
```

```
## 2291 samples
```

```
## 17 predictor
```

```
##
```

```
## No pre-processing
```

```
## Resampling: Cross-Validated (5 fold)
```

```
## Summary of sample sizes: 1833, 1833, 1833, 1831, 1834
```

```
## Resampling results across tuning parameters:
```

```
##
```

##	shrinkage	interaction.depth	n.minobsinnode	n.trees	RMSE	Rsquared
##	0.1	1	10	50	1.7367482	0.9706574
##	0.1	1	10	100	1.3437160	0.9795919
##	0.1	1	10	150	1.2599596	0.9818930
##	0.1	1	20	50	1.7336460	0.9709811
##	0.1	1	20	100	1.3270562	0.9800821
##	0.1	1	20	150	1.2468769	0.9822624
##	0.1	2	10	50	1.3483093	0.9800261
##	0.1	2	10	100	1.1635460	0.9845779
##	0.1	2	10	150	1.0856593	0.9865655
##	0.1	2	20	50	1.3440573	0.9800654
##	0.1	2	20	100	1.1668434	0.9844812
##	0.1	2	20	150	1.1032607	0.9861174
##	0.1	3	10	50	1.2228456	0.9832860
##	0.1	3	10	100	1.0508034	0.9874192
##	0.1	3	10	150	0.9641669	0.9894011
##	0.1	3	20	50	1.2126485	0.9835755
##	0.1	3	20	100	1.0453410	0.9875340
##	0.1	3	20	150	0.9749037	0.9891681

##	0.2	1	10	50	1.4584621	0.9758261
##	0.2	1	10	100	1.3361789	0.9796596
##	0.2	1	10	150	1.2851756	0.9811968
##	0.2	1	20	50	1.4355012	0.9766178
##	0.2	1	20	100	1.3182367	0.9802138
##	0.2	1	20	150	1.2669713	0.9817668
##	0.2	2	10	50	1.2345830	0.9826206
##	0.2	2	10	100	1.0977196	0.9862673
##	0.2	2	10	150	1.0283543	0.9879514
##	0.2	2	20	50	1.2376635	0.9825227
##	0.2	2	20	100	1.1111413	0.9859252
##	0.2	2	20	150	1.0421694	0.9876136
##	0.2	3	10	50	1.0898928	0.9864517
##	0.2	3	10	100	0.9661973	0.9893636
##	0.2	3	10	150	0.8898811	0.9909759
##	0.2	3	20	50	1.1203208	0.9857234
##	0.2	3	20	100	0.9912086	0.9888145
##	0.2	3	20	150	0.9141374	0.9904705
##	MAE					
##	1.3271193					
##	1.0555495					
##	0.9874441					
##	1.3133912					
##	1.0365122					
##	0.9722674					
##	1.0439934					
##	0.9042363					
##	0.8411656					
##	1.0459111					
##	0.9114520					
##	0.8580422					
##	0.9560513					
##	0.8169396					
##	0.7482082					
##	0.9371908					
##	0.8055222					
##	0.7467296					
##	1.1446434					
##	1.0451915					
##	1.0062149					
##	1.1201020					
##	1.0217979					
##	0.9834001					
##	0.9609237					
##	0.8486604					
##	0.7950935					
##	0.9619960					
##	0.8606764					
##	0.8007734					
##	0.8584652					
##	0.7562975					
##	0.6934345					
##	0.8727283					
##	0.7676555					

```
## 0.7028453
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were n.trees = 150, interaction.depth =
## 3, shrinkage = 0.2 and n.minobsinnode = 10.
```

```
print(time_ada)
```

```
## user system elapsed
## 3.786 0.061 3.895
```

Stacking algorithms In we perform ensemble in different way other than the standard approach where we consider three different weak models :

- Lasso
- Ridge
- Knn model

We wanted to combine two shrinkage algorithms to assess how these model perform compare to tree ensemble.

```
# ----- tree regression -----
time_tree <- system.time({
  tune_tree <- train(
    Life_expectancy ~ .,
    data      = training_dataExpectancy,
    method    = "rpart",
    trControl = ctrl,
    tuneGrid  = expand.grid(cp = c(0.001, 0.005, 0.01, 0.05, 0.1))
  )
})

# ----- Ridge -----
time_ridge <- system.time({
  set.seed(123)
  model_ridge <- train(
    Life_expectancy ~ .,
    data      = training_dataExpectancy,
    method    = "glmnet",
    trControl = ctrl,
    tuneGrid  = expand.grid(alpha = 0, lambda = 10^seq(-4, 2, length = 50)),
    preProcess = c("center", "scale")
  )
})

# ----- Lasso -----
time_lasso <- system.time({
  set.seed(123)
  model_lasso <- train(
    Life_expectancy ~ .,
    data      = training_dataExpectancy,
    method    = "glmnet",
    trControl = ctrl,
    tuneGrid  = expand.grid(alpha = 1, lambda = 10^seq(-4, 2, length = 50)),
```

```

    preProcess = c("center", "scale")
  )
})

```

```

## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.

```

```

# ---- KNN ----
time_knn <- system.time({
  set.seed(123)
  model_knn <- train(
    Life_expectancy ~ .,
    data      = training_dataExpectancy,
    method    = "knn",
    trControl = ctrl,
    tuneGrid  = expand.grid(k = c(3, 5, 7, 9, 11, 15)),
    preProcess = c("center", "scale")
  )
})

#

# ---- Stack meta-learner ----
time_stack <- system.time({
  pred_ridge <- predict(model_ridge, test_dataExpectancy)
  pred_lasso <- predict(model_lasso, test_dataExpectancy)
  pred_knn   <- predict(model_knn,   test_dataExpectancy)
  pred_tree  <- predict(tune_tree, test_dataExpectancy)
  meta_test  <- data.frame(
    ridge = pred_ridge,
    lasso = pred_lasso,
    knn   = pred_knn,
    tree  = pred_tree,
    Life_expectancy = test_dataExpectancy$Life_expectancy
  )
  meta_model <- lm(Life_expectancy ~ ridge + lasso + knn + tree, data = meta_test)
  pred_stack <- predict(meta_model, meta_test)
})

cat("Result of the prediction= \n")

```

```

## Result of the prediction=

```

```

print(head(pred_stack))

```

```

##      15      16      17      22      40      45
## 81.50504 79.53200 80.40544 68.82750 64.72741 78.09512

```

```
print(head(test_dataExpectancy$Life_expectancy))
```

```
## [1] 81.5 78.8 81.0 68.8 64.3 79.4
```

```
library(Metrics)
```

Evaluation of models

```
##  
## Attaching package: 'Metrics'  
  
## The following objects are masked from 'package:caret':  
##  
## precision, recall
```

```
actual <- test_dataExpectancy$Life_expectancy
```

```
pred_bag <- predict(model_bag$finalModel, test_dataExpectancy)  
pred_rf <- predict(model_rf$finalModel, test_dataExpectancy)  
pred_ada <- predict(model_ada$finalModel, test_dataExpectancy)
```

```
## Using 150 trees...
```

```
get_metrics <- function(actual, predicted, model_name, elapsed_time) {  
  tibble(  
    Model      = model_name,  
    RMSE       = round(rmse(actual, predicted), 4),  
    MAE        = round(mae(actual, predicted), 4),  
    Time_sec   = round(elapsed_time["elapsed"], 2)  
  )  
}  
  
final_table <- bind_rows(  
  get_metrics(actual, pred_bag, "Bagging", time_bag),  
  get_metrics(actual, pred_rf, "Random Forest", time_rf),  
  get_metrics(actual, pred_ada, "AdaBoost", time_ada),  
  get_metrics(actual, pred_ridge, "Ridge", time_ridge),  
  get_metrics(actual, pred_lasso, "Lasso", time_lasso),  
  get_metrics(actual, pred_knn, "KNN", time_knn),  
  get_metrics(actual, pred_tree, "tree", time_tree),  
  get_metrics(actual, pred_stack, "Stack (Ridge+Lasso+KNN)", time_stack)  
) %>% arrange(RMSE)  
  
print(final_table)
```

```
## # A tibble: 8 x 4  
##   Model      RMSE  MAE Time_sec  
##   <chr>      <dbl> <dbl>   <dbl>
```

```
## 1 Random Forest      0.490 0.360    64.1
## 2 Stack (Ridge+Lasso+KNN) 0.768 0.563    0.02
## 3 AdaBoost           0.891 0.703    3.89
## 4 KNN                 0.940 0.611    0.4
## 5 Lasso               1.37  1.08    0.12
## 6 Ridge               1.52  1.24    0.15
## 7 tree                1.68  1.34    0.1
## 8 Bagging             2.25  1.79    1.42
```

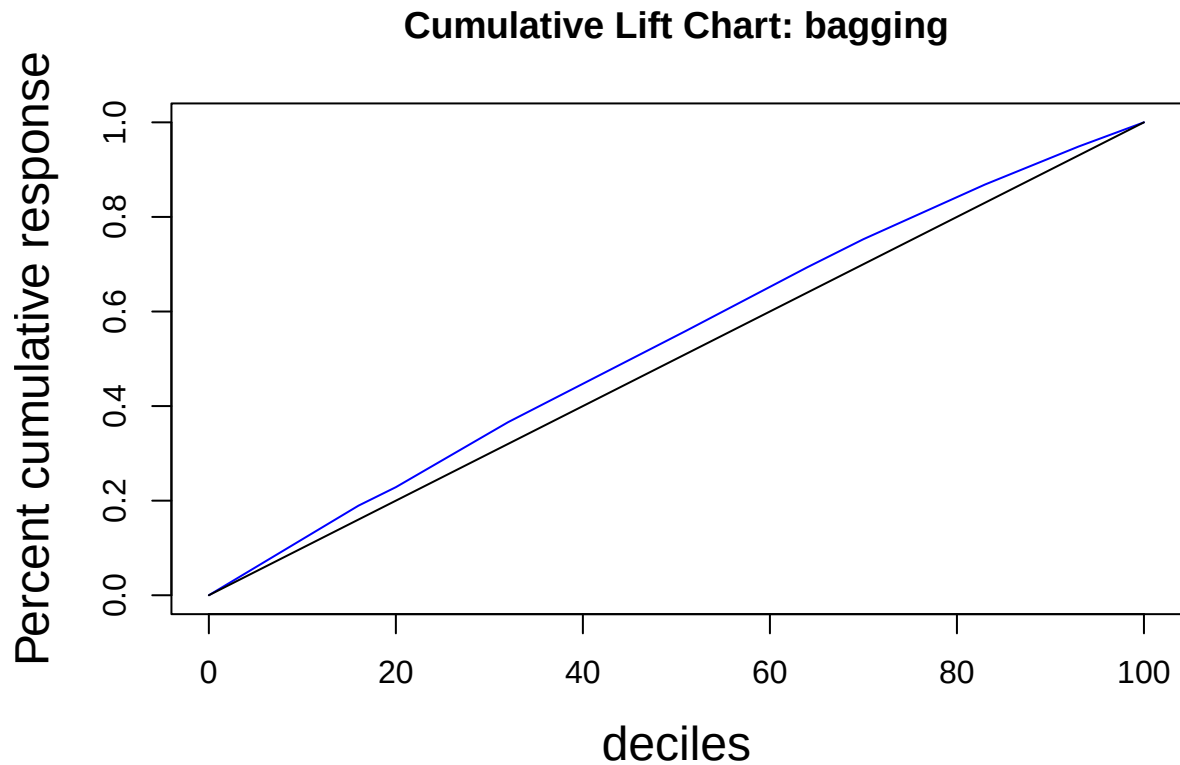
Lift plot We want to evaluate each model using the lift plot to understand the rate of contribution based on the number of observation collected.

- Bagging lift plot

```
# access the gain
gain_bag <- gains(test_dataExpectancy$Life_expectancy, pred_bag) # creates the cumulative gain chart
print(gain_bag)
```

## Depth				Cume	Cume Pct				Mean
## of		Cume	Mean	Mean	of Total	Lift	Cume	Model	
## File	N	N	Resp	Resp	Resp	Index	Lift	Score	
## 16	94	94	80.18	80.18	18.9%	115	115	79.75	
## 20	20	114	77.86	79.78	22.9%	112	115	78.95	
## 32	72	186	75.95	78.30	36.6%	109	113	75.53	
## 51	104	290	73.80	76.68	55.9%	106	110	73.06	
## 64	76	366	70.54	75.41	69.4%	102	109	70.62	
## 70	35	401	67.52	74.72	75.3%	97	108	68.34	
## 83	73	474	63.01	72.92	86.8%	91	105	62.59	
## 93	57	531	56.08	71.11	94.9%	81	102	56.08	
## 100	42	573	48.50	69.45	100.0%	70	100	50.14	
## NA	NA	NA	NA	NA	NA%	NA	NA	NA	

```
# Plot Lift chart: Percent cumulative response
x_bag <- c(0, gain_bag$depth)
pred.y_bag <- c(0, gain_bag$cume.pct.of.total)
avg.y_bag <- c(0, gain_bag$depth/100)
plot(x_bag, pred.y_bag, main = "Cumulative Lift Chart: bagging", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_bag, avg.y_bag, type = "l")
```



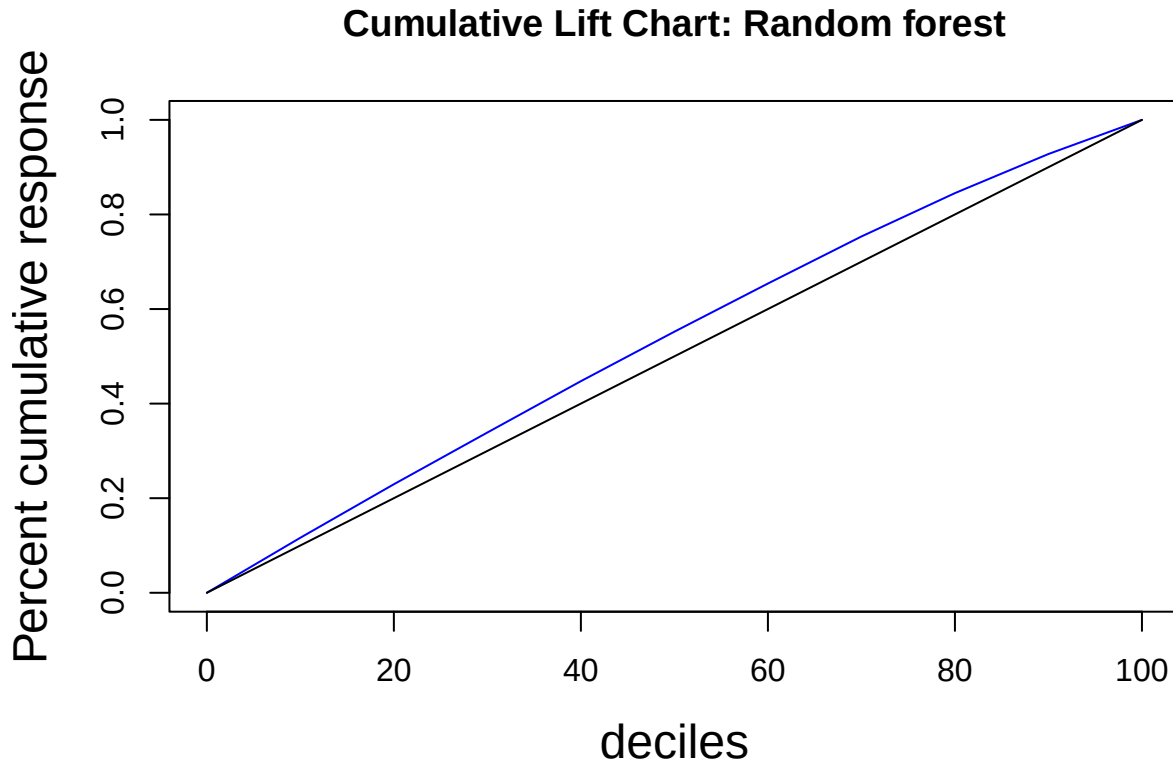
- Random Forest lift plot

```
# access the gain
gain_rf <- gains(test_dataExpectancy$Life_expectancy, pred_rf) # creates the cumulative gain chart
print(gain_rf)
```

## Depth				Cume	Cume Pct				Mean
## of		Cume	Mean	Mean	of Total	Lift	Cume		Model
## File	N	N	Resp	Resp	Resp	Index	Lift		Score
##	-----								
##	10	57	57	81.25	81.25	11.6%	117	117	81.15
##	20	57	114	79.00	80.12	23.0%	114	115	78.86
##	30	57	171	76.37	78.87	33.9%	110	114	76.30
##	40	58	229	74.46	77.75	44.7%	107	112	74.46
##	50	57	286	73.11	76.83	55.2%	105	111	73.11
##	60	57	343	71.10	75.88	65.4%	102	109	71.06
##	70	58	401	68.25	74.77	75.3%	98	108	68.21
##	80	57	458	63.99	73.43	84.5%	92	106	63.89
##	90	57	515	57.77	71.70	92.8%	83	103	57.79
##	100	58	573	49.52	69.45	100.0%	71	100	49.81

```
# Plot Lift chart: Percent cumulative response
x_rf <- c(0, gain_rf$depth)
pred.y_rf <- c(0, gain_rf$cume.pct.of.total)
avg.y_rf <- c(0, gain_rf$depth/100)
```

```
plot(x_rf, pred.y_rf, main = "Cumulative Lift Chart: Random forest", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_rf, avg.y_rf, type = "l")
```

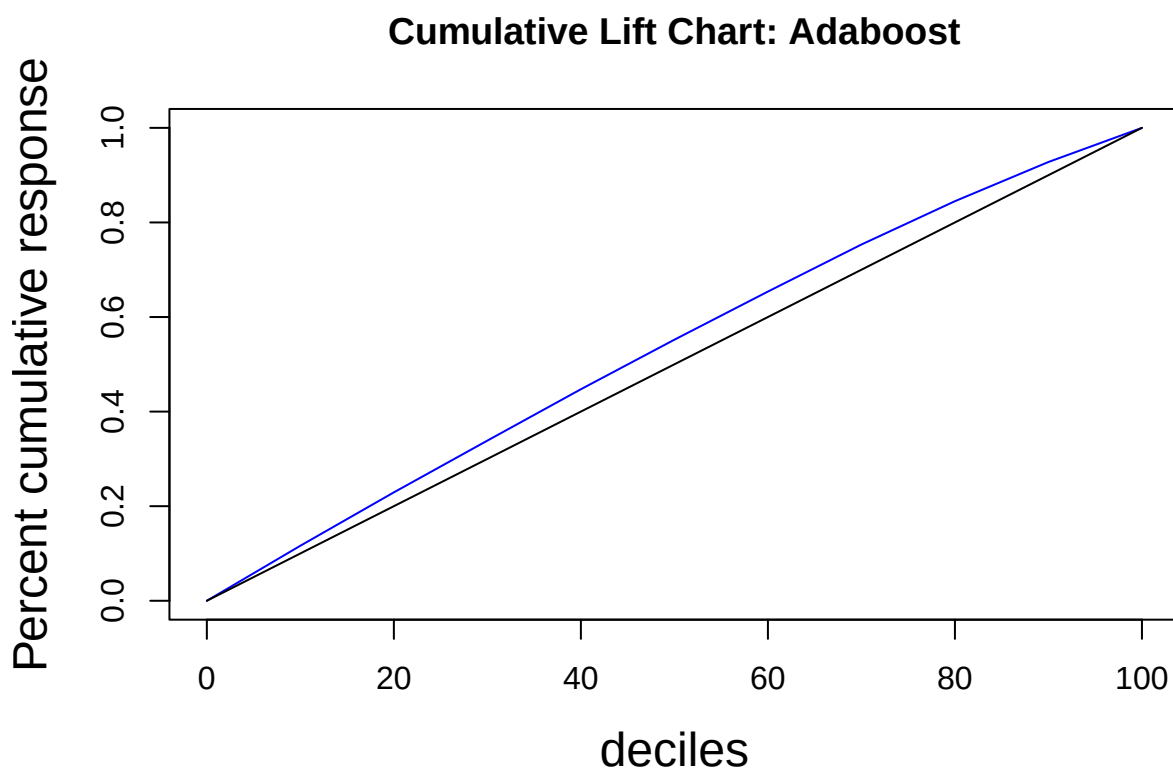


- Adaboost lift plot

```
# access the gain
gain_ada <- gains(test_dataExpectancy$Life_expectancy, pred_ada) # creates the cumulative gain chart
print(gain_ada)
```

##	Depth				Cume	Cume Pct			Mean
##	of		Cume	Mean	Mean	of Total	Lift	Cume	Model
##	File	N	N	Resp	Resp	Resp	Index	Lift	Score
##	10	57	57	81.19	81.19	11.6%	117	117	81.27
##	20	57	114	78.83	80.01	22.9%	114	115	78.88
##	30	57	171	76.43	78.82	33.9%	110	113	76.38
##	40	58	229	74.51	77.73	44.7%	107	112	74.49
##	50	57	286	73.17	76.82	55.2%	105	111	73.08
##	60	57	343	71.04	75.86	65.4%	102	109	70.94
##	70	58	401	68.26	74.76	75.3%	98	108	68.23
##	80	57	458	64.00	73.42	84.5%	92	106	63.97
##	90	57	515	57.84	71.70	92.8%	83	103	57.79
##	100	58	573	49.53	69.45	100.0%	71	100	49.61

```
# Plot Lift chart: Percent cumulative response
x_ada <- c(0, gain_ada$depth)
pred.y_ada <- c(0, gain_ada$cume.pct.of.total)
avg.y_ada <- c(0, gain_ada$depth/100)
plot(x_ada, pred.y_ada, main = "Cumulative Lift Chart: Adaboost", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_ada, avg.y_ada, type = "l")
```



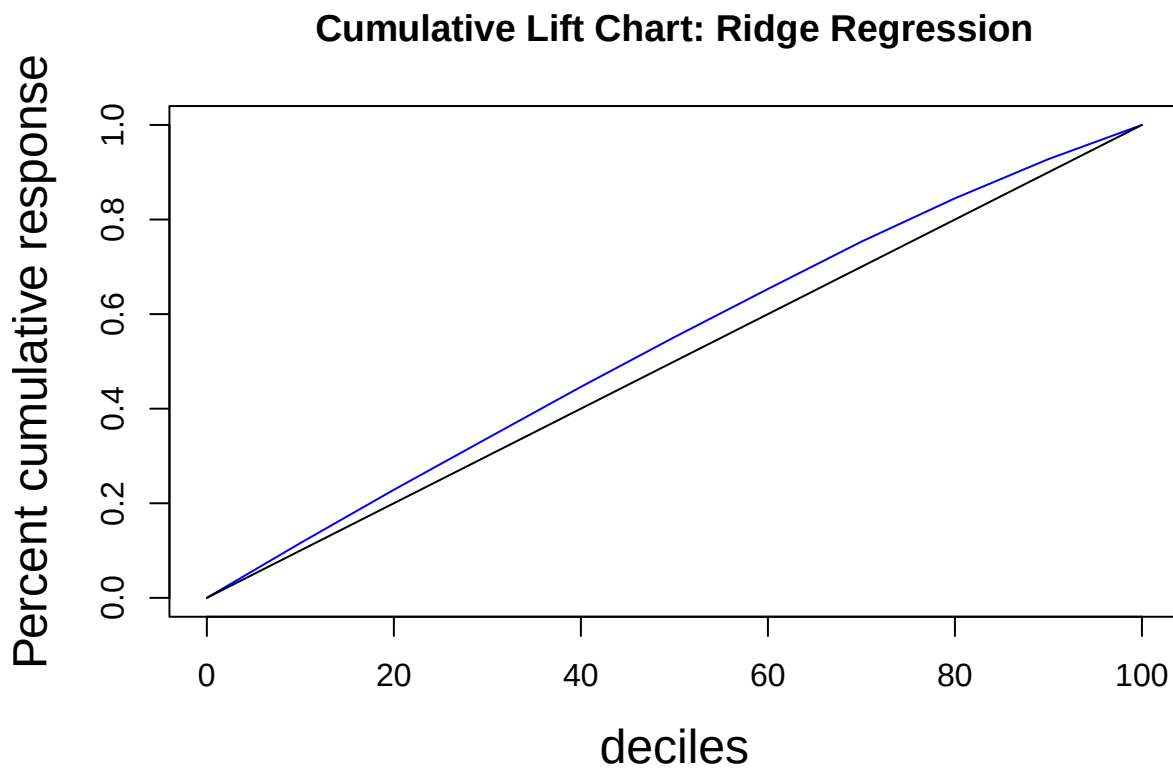
- Ridge Regression

```
# access the gain
gain_ridge <- gains(test_dataExpectancy$Life_expectancy, pred_ridge) # creates the cumulative gain char
print(gain_ridge)
```

##	Depth				Cume	Cume Pct			Mean
##	of		Cume	Mean	Mean	of Total	Lift	Cume	Model
##	File	N	N	Resp	Resp	Resp	Index	Lift	Score
##									
##	10	57	57	80.89	80.89	11.6%	116	116	80.38
##	20	57	114	78.69	79.79	22.9%	113	115	78.49
##	30	57	171	76.08	78.55	33.8%	110	113	75.69
##	40	58	229	74.54	77.54	44.6%	107	112	74.11
##	50	57	286	73.39	76.71	55.1%	106	110	72.82
##	60	57	343	71.12	75.78	65.3%	102	109	71.04

##	70	58	401	68.68	74.76	75.3%	99	108	68.44
##	80	57	458	63.98	73.41	84.5%	92	106	64.07
##	90	57	515	57.92	71.70	92.8%	83	103	58.85
##	100	58	573	49.52	69.45	100.0%	71	100	49.41

```
# Plot Lift chart: Percent cumulative response
x_ridge <- c(0, gain_ridge$depth)
pred.y_ridge <- c(0, gain_ridge$cume.pct.of.total)
avg.y_ridge <- c(0, gain_ridge$depth/100)
plot(x_ridge, pred.y_ridge, main = "Cumulative Lift Chart: Ridge Regression", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_ridge, avg.y_ridge, type = "l")
```



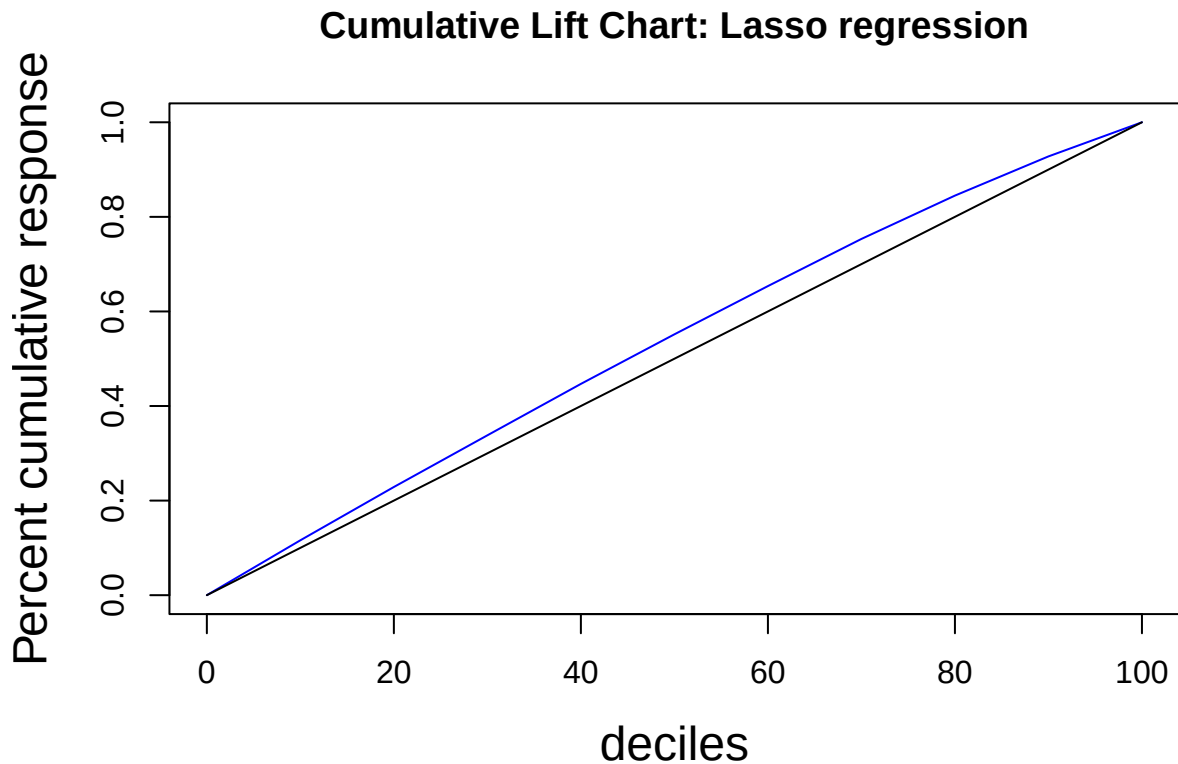
- Lasso Regression

```
# access the gain
gain_lasso <- gains(test_dataExpectancy$Life_expectancy, pred_lasso) # creates the cumulative gain char
print(gain_lasso)
```

##	Depth				Cume	Cume Pct			Mean
##	of		Cume	Mean	Mean	of Total	Lift	Cume	Model
##	File	N	N	Resp	Resp	Resp	Index	Lift	Score
##	-----								
##	10	57	57	81.01	81.01	11.6%	117	117	80.50

##	20	57	114	78.71	79.86	22.9%	113	115	78.72
##	30	57	171	76.29	78.67	33.8%	110	113	76.07
##	40	58	229	74.60	77.64	44.7%	107	112	74.37
##	50	57	286	73.20	76.75	55.2%	105	111	72.90
##	60	57	343	71.23	75.84	65.4%	103	109	71.09
##	70	58	401	68.39	74.76	75.3%	98	108	68.37
##	80	57	458	63.98	73.42	84.5%	92	106	64.06
##	90	57	515	57.87	71.70	92.8%	83	103	58.60
##	100	58	573	49.53	69.45	100.0%	71	100	48.81

```
# Plot Lift chart: Percent cumulative response
x_lasso <- c(0, gain_lasso$depth)
pred.y_lasso <- c(0, gain_lasso$cume.pct.of.total)
avg.y_lasso <- c(0, gain_lasso$depth/100)
plot(x_lasso, pred.y_lasso, main = "Cumulative Lift Chart: Lasso regression", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_lasso, avg.y_lasso, type = "l")
```



- Knn algorithm lift plot

```
# access the gain
gain_knn <- gains(test_dataExpectancy$Life_expectancy, pred_knn) # creates the cumulative gain chart
print(gain_knn)
```

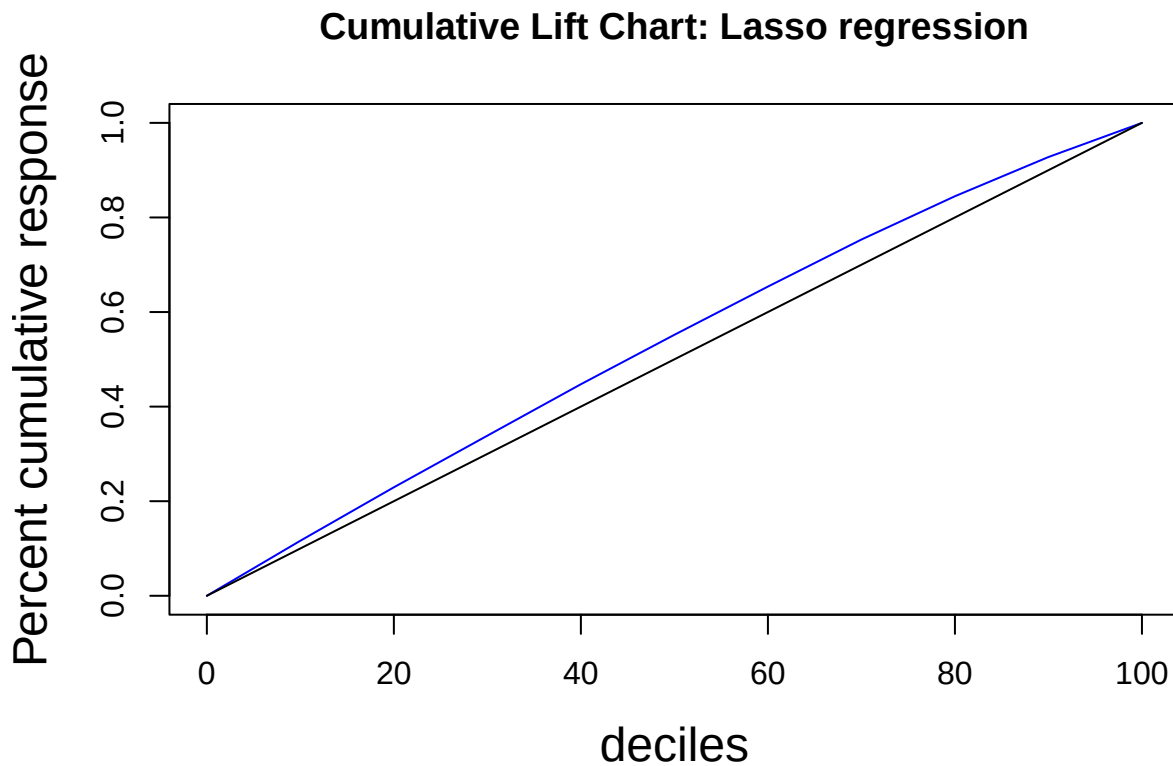
##	Depth	Cume	Cume Pct	Mean
----	-------	------	----------	------

##	of		Cume	Mean	Mean	of Total	Lift	Cume	Model
##	File	N	N	Resp	Resp	Resp	Index	Lift	Score
##	10	57	57	81.24	81.24	11.6%	117	117	81.10
##	20	57	114	78.92	80.08	22.9%	114	115	78.85
##	30	57	171	76.25	78.80	33.9%	110	113	76.30
##	40	58	229	74.59	77.74	44.7%	107	112	74.57
##	50	57	286	73.03	76.80	55.2%	105	111	73.05
##	60	57	343	71.17	75.86	65.4%	102	109	71.17
##	70	58	401	68.26	74.76	75.3%	98	108	68.16
##	80	57	458	63.92	73.41	84.5%	92	106	63.73
##	90	57	515	57.88	71.69	92.8%	83	103	58.06
##	100	58	573	49.56	69.45	100.0%	71	100	49.92

```

# Plot Lift chart: Percent cumulative response
x_knn <- c(0, gain_knn$depth)
pred.y_knn <- c(0, gain_knn$cume.pct.of.total)
avg.y_knn <- c(0, gain_knn$depth/100)
plot(x_knn, pred.y_knn, main = "Cumulative Lift Chart: Lasso regression", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_knn, avg.y_knn, type = "l")

```

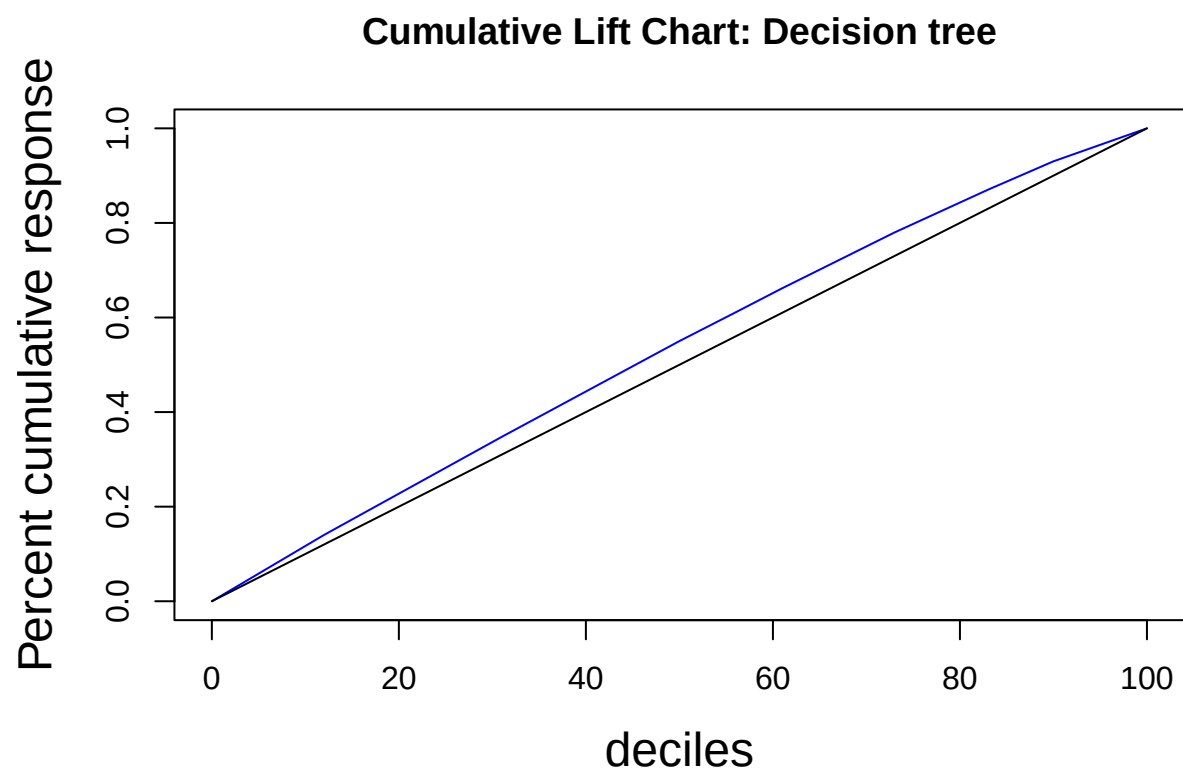


- tree regression

```
# access the gain
gain_tree <- gains(test_dataExpectancy$Life_expectancy, pred_tree) # creates the cumulative gain chart
print(gain_tree)
```

## Depth									
## of		Cume	Mean	Cume	Cume Pct			Mean	
## File	N	N	Resp	Mean	of Total	Lift	Cume	Model	
##				Resp	Resp	Index	Lift	Score	
##	12	69	69	80.89	80.89	14.0%	116	116	80.93
##	31	107	176	77.11	78.59	34.8%	111	113	76.92
##	50	109	285	73.94	76.81	55.0%	106	111	73.69
##	56	34	319	71.04	76.20	61.1%	102	110	71.06
##	61	29	348	70.13	75.69	66.2%	101	109	70.54
##	73	69	417	67.97	74.41	78.0%	98	107	67.77
##	83	58	475	62.12	72.91	87.0%	89	105	62.07
##	90	42	517	56.83	71.61	93.0%	82	103	57.05
##	100	56	573	49.56	69.45	100.0%	71	100	49.36
##	NA	NA	NA	NA	NA	NA%	NA	NA	NA

```
# Plot Lift chart: Percent cumulative response
x_tree <- c(0, gain_tree$depth)
pred.y_tree <- c(0, gain_tree$cume.pct.of.total)
avg.y_tree <- c(0, gain_tree$depth/100)
plot(x_tree, pred.y_tree, main = "Cumulative Lift Chart: Decision tree", xlab = "deciles",
     ylab = "Percent cumulative response", type = "l", col = "blue", cex.lab = 1.5)
lines(x_tree, avg.y_tree, type = "l")
```



- Decision tree

```
# plot the decision tree  
prp(tune_tree$finalModel, type = 1, extra = 1, under = TRUE, split.font = 2, varlen = -10)
```

